Performance Evaluation of AM Generation and Detection under Noisy Channel Conditions Using MATLAB

AIM :

- To generate an AM signal and add AWGN noise using MATLAB.
- To recover the message signal using envelope detection.
- To analyze how channel noise affects signal transmission and recovery.

CODE :

```
clc;

clear;

close all;


% Parameters

fs = 100000;

t = 0:1/fs:0.02;

fm = 500;

fc = 5000;

snr_db = 10;


% Message and carrier signals

m = cos(2*pi*fm*t);

c = cos(2*pi*fc*t);


% AM modulation

am = (1 + m).*cos(2*pi*fc*t);
```

```
% Add AWGN
signal_power = mean(am.^2);
noise_power = signal_power / (10^(snr_db/10));
noisy = am + sqrt(noise_power)*randn(size(am));

% Envelope detection
env = abs(hilbert(noisy));

% Plot signals
figure;

subplot(5,1,1);
plot(t,m);
title('Message Signal');
grid on;

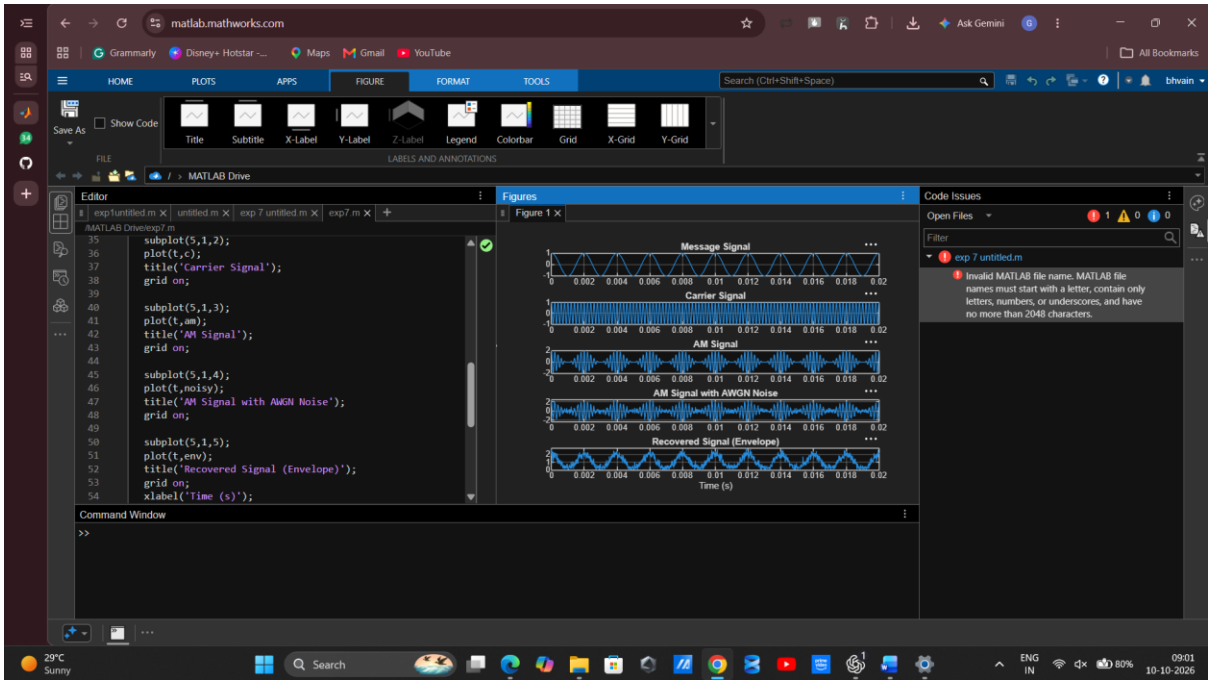
subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
grid on;

subplot(5,1,3);
plot(t,am);
title('AM Signal');
grid on;
```

```
subplot(5,1,4);
plot(t,noisy);
title('AM Signal with AWGN Noise');
grid on;

subplot(5,1,5);
plot(t,env);
title('Recovered Signal (Envelope)');
grid on;
xlabel('Time (s)');
```

OUTPUT:



RESULT :

- The AM signal was generated successfully, and the effect of AWGN was observed.
- The message signal was recovered using envelope detection.
- The plots showed how noise affected the transmitted signal and its recovery.